

provide accurate recommendations for planning future SK projects. This feature is designed to address the operational challenge of manual expenditure tracking and improve the transparency of SK financial operations.

Technical Background

The study utilizes a multi-layered software architecture consisting of React.js (v18.0) for the frontend and Node.js (v20.x) for server-side logic, integrated with a localized Large Language Model (LLM). React.js was selected for its superior component-based rendering, which is essential for maintaining a highly responsive mobile interface for SK officials. Instead of relying on traditional, monolithic API structures, the system leverages this integrated LLM to provide the specialized intelligence required for administrative tasks such as document automation and context-aware querying. The architecture is specifically optimized to run efficiently on accessible, mid-tier local hardware, eliminating the reliance on expensive enterprise cloud hosting or external API subscriptions. Tailwind CSS is utilized for UI design to ensure minimal load times on mobile devices, while VS Code serves as the primary IDE for developing the automated workflows and localized AI interactions.

Regarding hardware resources, the system follows a hardware-agnostic, web-based delivery model for the end-user, ensuring the framework is universally accessible via standard Android smartphones or tablets. To maximize flexibility and cost-efficiency for the Barangay, the system bypasses standard shared web-hosting environments—which

cannot sustain LLM workloads—in favor of self-hosting via secure internet tunneling. This tunneling method allows the beneficiary to utilize an existing local PC to expose the system to the internet, effectively deploying it as a fully functioning live web application without incurring recurring external server fees. This configuration is highly viable because the localized AI model is optimized to require significantly less capacity than larger enterprise alternatives, allowing the backend to operate stably on machines with 16GB of RAM. While the core data engineering and backend hosting are processed on a development machine equipped with an RTX 4060 Ti GPU, the final implementation remains entirely hardware-agnostic for the end-user. This enables SK officials to execute advanced, AI-driven administrative functions from their existing mobile devices without necessitating high-end equipment investments from the Barangay.